YASHASH RAVI

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AI/ML Engineer | Machine Learning | Deep Learning | NLP | Computer Vision

SUMMARY

Computer Science undergraduate specializing in Artificial Intelligence & Machine Learning with hands-on experience developing Machine Learning, Deep Learning, NLP, Computer Vision, and multi-agent AI systems. Experienced in building and integrating AI models for plant disease detection, facial emotion recognition, intelligent task automation, and information retrieval, with additional software engineering experience across 10+ full-stack applications. Strong foundation in Python, TensorFlow/Keras, scikit-learn, OpenCV, NLP, model evaluation, REST APIs, and production-oriented application development.

SKILLS

- **Programming:** Python, Java, C, JavaScript, SQL
- **Machine Learning:** Scikit-learn, Feature Engineering, Data Preprocessing, Classification, Model Evaluation, Random Forest, TF-IDF
- **Deep Learning:** PyTorch, TensorFlow, Keras, PyTorch Geometric (PyG), CNNs, RNNs, LSTM, GRU, BiGRU, GCN, Neural Networks, Transfer Learning
- **Graph Machine Learning:** Graph Neural Networks (GNN), Graph Convolutional Networks (GCN), Node Classification, Citation Networks, Graph Representation Learning, Message Passing
- **NLP:** Natural Language Processing, Text Classification, TF-IDF, Tokenization, Word Embeddings, Sequence Modeling, NLP Preprocessing, Hugging Face
- **Computer Vision:** OpenCV, Image Classification, Image Processing, CNNs
- **Generative AI & AI Agents:** Intent-Based Routing, Information Retrieval, Web Scraping, AI Task Automation
- **Model Deployment & APIs:** FastAPI, Flask, REST APIs, ONNX, ONNX Runtime, Pydantic, Model Serialization, Model Inference
- **Web Development:** React.js, Node.js, Express.js, Electron, REST API Integration
- **Databases:** PostgreSQL, MongoDB, SQL
- **Data Science & Visualization:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, NetworkX
- **Tools & DevOps:** Git, GitHub, Jupyter Notebook, Docker, Postman, Selenium, BeautifulSoup

EDUCATION

B.Tech. in Computer Science and Engineering - R.V. University — 2024 - 2028 (Expected)

- Specialization: Artificial Intelligence, Machine Learning and NLP (Natural Language Processing)
- Relevant Coursework: Data Structures & Algorithms, Probability & Statistics, Machine Learning, DBMS, Computer Networks, Operating Systems

WORK EXPERIENCE

Machine Learning Intern (Part-Time) | Tripple One Solutions — June 2025 - September 2025

Emotion-Based Music Recommendation System | Computer Vision, CNN, OpenCV, TensorFlow

- Developed a real-time facial emotion recognition system using computer vision and deep learning to classify emotional states from facial expressions.
- Worked on machine learning development and **AI-focused application** workflows, contributing to model development, data processing, evaluation, and integration.
- Applied Python-based machine learning techniques for data **preprocessing, model training, testing,** and **performance evaluation**.
- Contributed to integrating **ML functionality** into application workflows and supporting **end-to-end AI solution development.**

Web Development Intern (Part-Time) | Labmentix — April 2026 - July 2026

- Built and deployed **5+ full-stack web applications** using **React.js, Node.js, Express.js, MongoDB/PostgreSQL, REST APIs,** and modern web technologies.
- Developed backend modules for usage-based billing, API metering, Razorpay payments, **RBAC/JWT** authentication, accounting, inventory, ledger management, and real-time communication.
- Developed **20+ REST API** endpoints for API publishing, **subscriptions**, request metering, **usage tracking**, **billing analytics**, and **transaction management**.
- Contributed across the **SDLC**, including database design, **API development**, frontend integration, **debugging**, **testing**, **Git-based collaboration**, and **deployment**.

PROJECTS

Graph Neural Network (GNN) | Research Paper Topic Classification (PyTorch Geometric | FastAPI | ONNX) — [LIVE DEMO] [Pipeline] [Project repository]

- Built an end-to-end Graph Neural Network pipeline for citation-based research paper classification using the Cora dataset (**2,708** papers, **10,556** citation edges, **1,433-dimensional node features**, **7 topic classes**); implemented a **2-layer GCN** with **GCNConv, ReLU, Dropout, Adam optimize**r, Cross-Entropy Loss, validation-based early stopping, and checkpointing.
- Developed and benchmarked a **TF-IDF + Random Forest baseline** against the **GCN model,** achieving **58.00%** → **78.20%** accuracy (**+20.20** percentage points) and **58.18%** → **77.74% Macro F1** (**+19.56** points) through graph-based message passing that combines textual node features with citation relationships.
- Exported the trained **PyTorch Geometric GCN** model to ONNX (Opset 18) with dynamic node/edge dimensions and integrated ONNX Runtime for CPU inference; implemented FastAPI REST endpoints for custom graph prediction, real Cora node classification, model metadata, health monitoring, input validation, probability/logit generation, and 7-class topic prediction.
- Designed a deployable inference layer using **FastAPI + Pydantic + NumPy + ONNX Runtime**, supporting arbitrary graph inputs and Cora nodes with softmax confidence probabilities, predicted class labels, **logits**, node/edge counts, and static frontend serving, enabling production-oriented cross-platform GNN inference.

[Pipeline]

- Cora Dataset → Graph Preprocessing → TF-IDF + Random Forest Baseline → 2-Layer GCN → Training & Early Stopping → Accuracy/Macro-F1 Evaluation → PyTorch → ONNX Export → ONNX Runtime → FastAPI REST API → Research Topic Prediction

Emotion Classification using RNN, LSTM, GRU & BiGRU | NLP | TensorFlow/Keras

GitHub repository

- Developed an end-to-end NLP emotion classification system using the Hugging Face dair-ai/emotion dataset containing **20,000 labeled** text samples across **6 emotion** classes (joy, sadness, anger, fear, love, surprise), with balanced class-weight handling for imbalanced training data.
- Built and compared Simple **RNN, LSTM, GRU**, and **Bidirectional GRU (BiGRU)** architectures using an Embedding → Recurrent Layers → Dropout → Softmax pipeline; implemented **10,000-word vocabulary**, **50-token padded sequences**, **128-dimensional embeddings**, and **0.5 dropout** for sequence modeling.
- Achieved **92.00%** test accuracy with BiGRU and 0.2099 test loss, outperforming **GRU** (**34.45%**), **LSTM** (**11.15%**), and **Simple RNN** (**10.30%**); achieved **+57.55 percentage** points over GRU and **+81.70 points** over Simple RNN in test accuracy.
- Implemented model training with **Adam optimizer**, **sparse categorical cross-entropy**, **balanced class weights**, **early stopping (patience=3)**, and best-weight restoration; generated confusion-matrix analysis and sample predictions, and saved the trained **BiGRU Keras model** (.keras) and **tokenizer** (.pkl) for inference.

Plant Disease Detection | CNN, TensorFlow, Keras, OpenCV, Flask, React

GitHub repository

- Developed an end-to-end CNN-based plant disease detection system that processes leaf images through an **OpenCV** preprocessing pipeline and classifies **11+ plant** health/disease categories using a **TensorFlow/Keras model.**
- Built a **Flask REST API** inference pipeline for image **decoding, resizing, tensor preparation, model inference, confidence scoring**, and **60% confidence-threshold** validation, integrated with a React/Vite frontend.
- Engineered an interactive **AI-powered diagnosis** interface with **drag-and-drop image upload**, **disease-specific treatment** and **care recommendations**, prediction history, and responsive visualization for real-world agricultural use.

Pentrilon | Multi-Agent AI Desktop Assistant (Flask | PERN Stack | Electron)

GitHub repository

- Built a cross-platform AI desktop assistant integrating **Electron, React, Redux Toolkit,** and **Flask** for **natural-language** and **voice-driven desktop interaction**.
- Developed the IntrusFormat instruction engine to convert natural-language commands into validated JSON instruction schemas, enabling structured and controlled execution of **multi-step desktop tasks**.
- Integrated **Whisper, Selenium, AppOpener, PyAutoGUI**, and **Psutil** for **speech recognition**, browser/application automation, and **real-time CPU**, **RAM,** and **battery monitoring**

ADDITIONAL SOFTWARE PROJECTS

Live Social Media Application | PERN, Cloudinary, Socket.IO

Live Demo | GitHub repository

- Built a full-stack social networking platform with posts, bookmarks, threaded comments, profiles, media sharing, communities, and real-time messaging.
- Implemented secure JWT authentication, authorization, REST APIs, PostgreSQL integration, one-to-one/group messaging, and Socket.IO-based real-time communication.
- Integrated Cloudinary for media management and Firebase for push notifications with a responsive React/Vite frontend.